

Green Favoritism: Leader Birth Regions Receive Cleaner Growth

Richard Schulz^{a,b}, Muhammad Majid Altaf^a, and Ruby Wang^a

^aUniversity of California, Berkeley, Berkeley, CA 94720, USA; ^bETH Zurich, 8092 Zurich, Switzerland

This manuscript was compiled on May 4, 2026

We replicate the birth-region nightlights favoritism documented by Bora (2025) and Düben, Hodler, and Raschky (2026) and extend the result to air pollution. We define *green favoritism* as a negative birth-region coefficient on pollution intensity, i.e. log pollution minus log nightlights, meaning that birth regions grow brighter without a proportional increase in air pollution. Using a harmonized DMSP-to-VIIRS panel for 2005–2017 (622,868 district-year observations), the birth-region coefficient on log nightlights is 0.049 (SE = 0.015). NO₂ intensity falls by 0.039 (SE = 0.019) and PM_{2.5} intensity by 0.044 (SE = 0.015) at leader entry. An event study with 120 balanced spells confirms flat NO₂ pre-trends and a negative pollution-intensity effect peaking at $t + 2$. The effect is concentrated in Sub-Saharan Africa, where both the activity and the green favoritism coefficients are roughly three times the global estimate. Globally, the activity effect is also about three times larger in mineral-rent economies, consistent with a composition channel in which leaders of resource-rich states direct large public works with low pollution intensity toward home districts.

regional favoritism | political economy | nighttime lights | air pollution
| green favoritism

Political leaders direct resources toward their birth regions. Cross-country panels of subnational nightlights have documented this pattern repeatedly: districts that are home to the sitting national leader grow brighter relative to comparable districts in the same country-year (1–3). A natural follow-up question, however, has not been examined: *do national leaders' birth regions receive disproportionately clean economic growth?* If leaders channel public goods and infrastructure toward home districts, pollution should rise less than economic activity. If they route industrial subsidies or manufacturing instead, pollution should rise proportionally or more.

We find that birth-region growth is clean. Using a harmonized DMSP-to-VIIRS panel for 2005–2017, we first replicate the nightlights result with a birth-region coefficient of 0.049 (SE = 0.015, $p = 0.001$). Both NO₂ intensity and PM_{2.5} intensity—log pollution minus log nightlights, measuring pollution per unit of proxied economic activity—are significantly negative: -0.039 (SE = 0.019) and -0.044 (SE = 0.015) respectively. Birth regions grow brighter without a proportional increase in air pollution. We term this result *green favoritism*.

An event study with a balanced ± 3 -year window around leader entry (120 spells, entry 2008–2016) provides the timing evidence and the parallel-trends test. Pre-trends for NO₂ intensity are flat at $t = -3$ and $t = -2$ (neither coefficient is statistically significant), supporting the parallel-trends assumption for the pollution outcome. NTL and PM_{2.5} pre-trends are significant, consistent with the observation that leaders tend to come from economically active regions. The pollution-intensity effect turns negative at $t + 2$ (-0.110^*) and fades by $t + 3$, indicating a short-run effect concentrated in the first two to three years of a spell.

Hodler and Raschky (2014) establish birth-region favoritism using DMSP nightlights. Bora (2025) and Düben, Hodler, and Raschky (2026) confirm the result with updated birthplace data and extended samples. De Luca et al. (2018) extend the favoritism result to ethnic homelands (4), and Bompreszi et al. (2024) document a parallel pattern for the spousal birthplaces of leaders (5). The literature on regional favoritism is, in this sense, well populated on the side of *how much* activity birth regions receive, but largely silent on the *kind* of activity they receive.

A separate literature studies the political economy of pollution. Cross-country work links environmental outcomes to trade openness, factor endowments, and the strength of regulatory institutions (6). Our paper sits at the intersection of this literature and the favoritism literature: the favoritism literature has examined how much activity birth regions receive but not what kind, while the pollution literature has examined cross-country and within-country drivers of emissions but not the role of leader birthplace. The contribution here is to show that political connection shapes not only the level of activity but its environmental composition. The pollution-intensity result is also independent of the nightlights result in the sense that air-quality satellites measure a different physical quantity from luminance, and the intensity outcome differences out the nightlights component, so it cannot be a mechanical artifact of the favoritism documented in prior work.

This paper documents green favoritism, a pollution-intensity result that does not follow mechanically from the nightlights finding and that uses independent satellite data on

Significance Statement

Birth regions of national leaders grow brighter while in office, and they grow cleaner. Using a harmonized 2005–2017 panel, we find a nightlights coefficient of 0.049 ($p = 0.001$), an NO₂ intensity coefficient of -0.039 ($p = 0.044$), and a PM_{2.5} intensity coefficient of -0.044 ($p = 0.004$). Birth regions receive economic activity that does not come with proportional air pollution. An event study with 120 balanced spells shows flat NO₂ pre-trends and a negative pollution-intensity effect at $t + 2$. The effect is concentrated in Sub-Saharan Africa, and globally the activity coefficient is about three times larger in mineral-rent economies, consistent with a composition channel in which leaders direct large public works funded by resource rents toward birth regions.

R.S. conceived the study, built and integrated the main data pipeline, ran the core empirical analysis, and drafted the manuscript. M.M.A. contributed to the NO₂ analysis. R.W. contributed to manuscript review and proofreading.

The authors declare no competing interests.

To whom correspondence should be addressed. E-mail: rschul@ethz.ch

air quality. The effect is concentrated in Sub-Saharan Africa, where both activity and pollution-intensity effects are roughly three times the global pooled estimate.

Section 1 describes the data. Section 2 presents the replication. Section 3 documents green favoritism. Section 4 turns to the event study. Section 5 examines geographic heterogeneity, and Section 6 discusses mechanisms.

1. Data

The analysis combines five core inputs. First, the Political Leaders' Affiliation Database (PLAD) provides leader identity, spell timing, and subnational birthplace information used to code treatment (7). Second, nightlights provide the economic-activity proxy. For replication we use a DMSP ADM2 panel for 1992–2013. For the main analysis we use a harmonized DMSP-to-VIIRS panel for 2005–2017. DMSP records luminance as an integer Digital Number (DN) from 0 to 63, with known saturation near the ceiling (8). Our zonal statistics pipeline treats zero-valued pixels as valid dark observations rather than missing data, preserving the extensive margin of electrification. Third, we use ACAG surface NO₂ data for 2005–2017 (9). Fourth, we use ACAG SatPM PM_{2.5} as a complementary pollution measure. Fifth, GADM ADM2 boundaries provide the common subnational geography (10).

Raster extraction follows a uniform pipeline across all satellite products. Annual rasters are aggregated to ADM2 polygons by computing the mean of all valid pixels intersecting each boundary polygon (zonal statistics). Zero-valued pixels (districts with no detectable nightlights or surface concentrations) are treated as valid observations rather than missing data, which preserves the extensive margin of electrification and retains variation across dark, low-pollution regions. Only pixels designated `nodata` by the source product (ocean, ice, and out-of-bounds cells) are excluded from the mean. Outcome variables enter regressions as $\log(Y_{ict} + 0.01)$; the constant shift avoids undefined logarithms for zero-valued observations and maps dark districts to approximately -4.6 . Results are robust to alternative shift values of 0.001 and 1. Pollution intensity is $\log(\text{Pollution} + 0.01) - \log(\text{Nightlights} + 0.01)$, measuring emission output per unit of proxied economic activity.

Treatment is defined at the district-year level. A district is treated when it is the birth region of the national leader in office in that year. To ensure a unique treatment assignment in transition years, we truncate the earlier of any two consecutive leader spells that overlap within a country.

The core estimating equation uses ADM2 fixed effects and country-year fixed effects:

$$\log(Y_{ict} + 0.01) = \beta \text{BirthRegion}_{ict} + \alpha_i + \gamma_{ct} + \varepsilon_{ict},$$

where i indexes ADM2 regions, c countries, and t years. The ADM2 fixed effects α_i absorb all time-invariant district characteristics: geography, historical institutions, and long-run income. The country-year fixed effects γ_{ct} absorb national policy shocks, macroeconomic cycles, and any aggregate trend in pollution or nightlights common to all districts of a country in a given year. The coefficient β is therefore identified by within-country-year variation: whether a district that becomes the current leader's birth region experiences differential growth relative to other districts in the same country at the same time.

Table 1. Summary statistics, 2005–2017 analysis sample

Variable	Mean	SD	N
Log nightlights (harmonized)	0.508	2.351	622,868
NO ₂ (ppb)	0.972	2.079	585,141
PM _{2.5} (μg/m ³)	18.051	11.472	620,805
Birth-region indicator	0.003	0.057	622,868

Unit of observation is ADM2 district-year. The birth-region indicator equals one when the district is the birth region of the sitting national leader. Observations differ across variables due to differences in satellite coverage.

Table 2. Replication ladder: birth-region nightlights effect across sensor generations

Step	Outcome	Coef.	SE	<i>p</i>
(1) DMSP 1992–13	Log NTL	0.043	0.014	0.002
(2) DMSP 2005–13	Log NTL	0.033	0.014	0.019
(3) Harm. 2005–17	Log NTL	0.049	0.015	0.001

Lag-0 birth-region coefficients on log nightlights across sensor generations ($N = 1,054,103$; 431,208; 622,868). All rows: ADM2 FE, country-year FE, spell-clustered SE. Row (3) is the specification carried forward to the pollution outcomes in Table 3.

The main threat to identification is pre-existing trends. Leaders may disproportionately originate from districts already on different growth trajectories, so that the post-treatment levels reflect pre-existing conditions rather than causal favoritism. This is the selection concern raised by Hodler and Raschky (2014), and the event study in Section 4 tests it directly by examining pre-period coefficients. The pollution-intensity outcome carries an additional advantage: because it differences out the nightlights component, common drivers of both nightlights and pollution (including most pre-existing economic trends) cancel. This makes the green favoritism coefficient harder to generate spuriously than the nightlights coefficient alone.

Standard errors are clustered by leader spell in the baseline, following the convention in recent replication work (2). Pollution intensity is $\log(\text{Pollution} + 0.01) - \log(\text{Nightlights} + 0.01)$; because it uses the same regressor and fixed effects as the component regressions, its coefficient is mechanically $\beta_{\text{pollution}} - \beta_{\text{NTL}}$. A negative intensity coefficient means pollution rises less than nightlights in birth regions.

Table 1 reports summary statistics for the 2005–2017 analysis sample.

2. Replication

Table 2 presents a replication ladder for the birth-region nightlights effect across successive sample windows and sensor generations. In the long DMSP 1992–2013 window the estimate is 0.043 ($p = 0.002$); in the overlap DMSP 2005–2013 window, 0.033 ($p = 0.019$); in the harmonized 2005–2017 panel, 0.049 ($p = 0.001$). The estimates are of similar magnitude and all statistically significant across sensor generations, consistent with Bora (2025) and Düben, Hodler, and Raschky (2026). The harmonized 2005–2017 specification in row (3) is the one we extend to pollution outcomes in Section 3.

Table 3. Green favoritism: birth regions grow brighter and cleaner (2005–2017)

	(1) ln NTL	(2) NO ₂ int.	(3) PM _{2.5} int.
Birth-region indicator	0.049*** (0.015)	−0.039* (0.019)	−0.044** (0.015)
<i>N</i>	622,868	585,141	620,805
Region FE	Yes	Yes	Yes
Country-year FE	Yes	Yes	Yes
SE clustering	Spell	Spell	Spell

Pollution intensity is $\log(\text{Pollution} + 0.01) - \log(\text{Nightlights} + 0.01)$. Negative intensity coefficients mean pollution rises less than nightlights in birth regions. Observations differ across columns because NO₂ and PM_{2.5} have different spatial coverage. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Standard errors clustered by leader spell.

3. Green Favoritism

Table 3 presents the main result. In the harmonized 2005–2017 panel, the birth-region coefficient on log nightlights is 0.049 (SE = 0.015, $p = 0.001$), directly replicating the result in Table 2. The new result is the pollution side: the birth-region coefficient on NO₂ intensity is −0.039 (SE = 0.019, $p = 0.044$) and on PM_{2.5} intensity is −0.044 (SE = 0.015, $p = 0.004$). Both pollution-intensity coefficients are negative and statistically significant. Birth regions receive economic activity that does not come with proportional air pollution.

The green favoritism result does not require that pollution falls in absolute terms. The lag-0 NO₂ level coefficient is small and insignificant at 0.006 ($p = 0.585$). The negative intensity is driven by a large positive nightlights coefficient combined with a near-zero pollution response: birth regions grow brighter without growing proportionally dirtier.

The PM_{2.5} result is more robust to clustering than the NO₂ result. Under country-level clustering (Table 8), nightlights and PM_{2.5} intensity remain significant while NO₂ intensity loses precision. Both indicators point in the same direction; PM_{2.5} captures a broader mix of emission sources including transportation and combustion, while NO₂ is more specific to traffic and industrial combustion.

Table 4 shows that the effects persist across lags. The nightlights effect peaks at lag 1 (+0.063, $p < 0.001$), consistent with a one-year delay between fiscal allocation and measurable luminosity. Both pollution-intensity effects remain negative and statistically significant through lag 2, indicating that the clean-composition advantage is not transitory.

4. Event Study

Figure 1 shows the event study for log nightlights, NO₂ intensity, and PM_{2.5} intensity. The sample is a balanced ± 3 -year window around leader entry, restricted to spells entering 2008–2016; the post-window for later entries extends beyond 2017, so this analysis uses the harmonized panel through 2019 solely to preserve balance. This yields 120 balanced spells. The reference period is $t = -1$; the solid red line marks entry ($t = 0$) and the dashed red line marks exit.

The NO₂ intensity panel provides the strongest parallel-trends evidence. Pre-trend coefficients at $t = -3$ and $t = -2$ are close to zero, and neither is statistically significant, supporting parallel trends for the pollution outcome. The effect turns negative at $t + 2$ (−0.110*) and fades to near zero

Table 4. Green favoritism across lags (2005–2017)

	(1) ln NTL	(2) NO ₂ int.	(3) PM _{2.5} int.
Lag 0	0.049*** (0.015)	−0.039* (0.019)	−0.044** (0.015)
Lag 1	0.063*** (0.015)	−0.081*** (0.019)	−0.059*** (0.015)
Lag 2	0.042** (0.016)	−0.070*** (0.021)	−0.039* (0.016)
<i>N</i>	622,868	585,141	620,805
Region FE	Yes	Yes	Yes
Country-year FE	Yes	Yes	Yes
SE clustering	Spell	Spell	Spell

Each cell is a separate regression. Lag k uses the birth-region indicator shifted k years forward. All rows use harmonized 2005–2017 panel, ADM2 fixed effects, country-year fixed effects, and leader-spell clustered standard errors. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 5. Geographic decomposition: 2005–2017 nightlights

Sample	Coef.	SE	<i>p</i> -value	<i>N</i>
Pooled	0.049	0.015	0.001	622,868
Sub-Saharan Africa	0.109	0.046	0.017	56,589
Non-SSA	0.033	0.015	0.026	566,279
Africa	0.101	0.040	0.012	85,063
Low NO ₂ dispersion	0.108	0.033	0.001	92,499

Lag-0 birth-region coefficients on log nightlights, harmonized 2005–2017. All rows use ADM2 fixed effects, country-year fixed effects, and leader-spell clustered standard errors. Low-NO₂-dispersion countries are defined as countries below the median cross-regional standard deviation of NO₂ in 2005–2007.

by $t + 3$. The NTL and PM_{2.5} panels show significant pre-trends at $t = -3$ and $t = -2$: birth regions were already on a different trajectory before the leader took office. These pre-trends are not a specification problem; they are expected given that leaders tend to come from economically active regions.

The $t + 3$ fade in all three panels, even with perfectly balanced composition at that horizon, suggests the favoritism effect is short-run rather than a permanent change in district trajectory. Whether this reflects project completion, political cycles, or reduced marginal returns to home-region spending remains an open question.

5. Geographic Heterogeneity

The green favoritism result is concentrated in Sub-Saharan Africa. Table 5 decomposes the 2005–2017 nightlights effect by region. The SSA coefficient is 0.109 ($p = 0.017$, $N = 56,589$), roughly three times the pooled estimate of 0.049. The non-SSA estimate is 0.033 ($p = 0.026$, $N = 566,279$). Both are significant, but the SSA magnitude is substantially larger. The SSA pollution-intensity estimates are also larger in magnitude: NO₂ intensity at −0.106 (SE = 0.051, $p = 0.042$) and PM_{2.5} intensity at −0.112 (SE = 0.044, $p = 0.013$).

The SSA concentration is consistent across replication windows: the SSA coefficient is 0.125 in DMSP 1992–2013, 0.116 in DMSP 2005–2013, and 0.109 in harmonized 2005–2017, always roughly three times the pooled estimate.

6. Mechanism

The green favoritism pattern is consistent with leaders channeling investment with low emission intensity toward birth

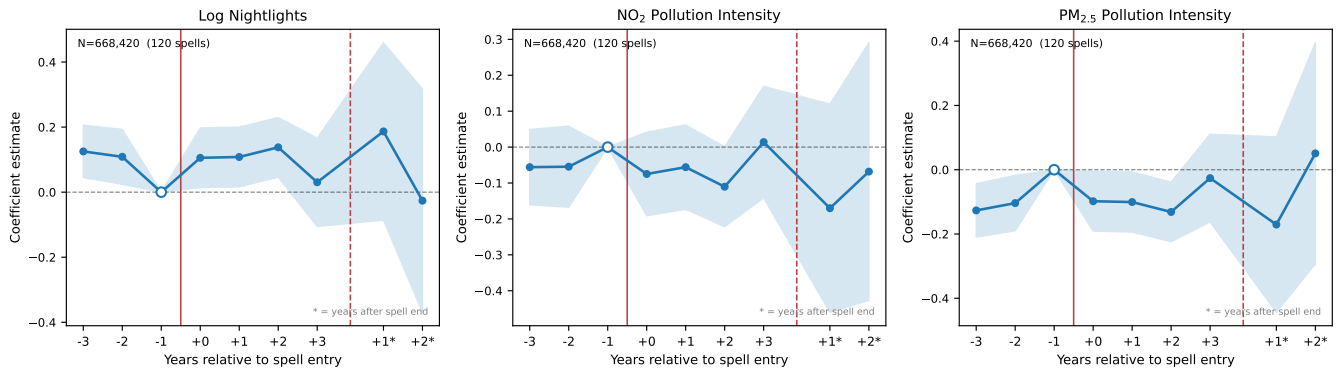


Fig. 1. Event study around leader entry, balanced ± 3 -year window (120 spells, entry 2008–2016). Three panels show log nightlights, NO_2 intensity, and $\text{PM}_{2.5}$ intensity. Coefficients are relative to $t = -1$; solid red line = entry ($t = 0$), dashed red line = exit. Bars are 95% confidence intervals. NO_2 pre-trends at $t = -3$ and $t = -2$ are flat; NTL and $\text{PM}_{2.5}$ pre-trends are significant, consistent with leaders coming from already economically active regions.

regions, such as large public works (ports, airports, government facilities) that generate nightlights without proportional industrial emissions. We cannot directly test project composition with the available data, but the case of Sri Lanka under President Mahinda Rajapaksa (2005–2015) is illustrative. Rajapaksa was born in Hambantota District; during his presidency, the government constructed the Hambantota International Port, Mattala Rajapaksa International Airport, and a major convention center in that district, infrastructure projects that substantially increased local luminosity without industrial activity. Whether this pattern generalizes across countries requires project-level investment data that we do not have.

Three channels could generate negative pollution intensity in birth regions. The first is a *composition channel*: leaders direct low-emission investment (ports, airports, roads, government facilities) toward birth regions instead of heavy manufacturing or industrial plants. The Hambantota case fits this channel: the flagship projects are transport and logistics infrastructure, not industrial production. The second is an *enforcement channel*: birth-region firms may face stricter environmental enforcement while the leader is in office, reducing emissions per unit of output. The third is a *sorting channel*: birth regions may attract already-clean sectors (services, tourism, knowledge industries) rather than manufacturing. We cannot distinguish these channels with aggregate satellite data. However, the democracy heterogeneity (Section 6) bears on the enforcement channel: if enforcement drove green favoritism, one would expect a stronger effect in democratic states where monitoring capacity tends to be higher. The concentration in low-democracy countries is instead more consistent with the composition channel, where political discretion over project allocation is greater when accountability is weaker.

Mineral rents and the composition channel. If the composition channel is correct, the favoritism effect should be stronger in countries with the fiscal slack and political discretion to fund large public-works projects from resource revenues. Large transport and energy infrastructure in resource-rich states (the Hambantota case is one example among many) is precisely the kind of investment that raises nightlights without proportional

Table 6. Mineral-rent decomposition (DMSP 1992–2013)

Sample	Coef.	SE	p
All countries	0.042	0.014	0.003
excl. top-quartile mineral	0.028	0.015	0.061
top-quartile mineral only	0.082	0.034	0.017
Sub-Saharan Africa	0.125	0.031	0.000
excl. top-quartile mineral	0.116	0.041	0.005
top-quartile mineral only	0.139	0.047	0.003
Non-SSA, top-quartile mineral	0.048	0.047	0.314
Non-SSA, excl. top-quartile mineral	0.009	0.016	0.553

Lag-0 birth-region coefficients on log nightlights, DMSP 1992–2013. All rows: ADM2 fixed effects, country-year fixed effects, and leader-spell clustered standard errors. Mineral-rent quartiles are constructed from country-mean WDI mineral rents (% of GDP) over 1992–2013.

industrial emissions. We therefore test whether birth-region favoritism varies with exposure to mineral rents.

Table 6 splits the long DMSP 1992–2013 sample by whether a country falls in the top quartile of country-mean mineral rents (% of GDP, World Bank WDI). The long DMSP window is used here because it gives the mineral-rent split adequate statistical power; the same exercise on the harmonized 2005–2017 window points in the same direction but with wider confidence intervals. On the WDI-merged sample, the global activity coefficient is 0.042 ($p = 0.003$). Excluding the top-quartile mineral-rent countries lowers it to 0.028 ($p = 0.061$), while restricting to that quartile alone raises it to 0.082 ($p = 0.017$). Mineral-rent economies thus exhibit a roughly three-fold-larger birth-region effect than non-mineral economies, consistent with a composition channel funded by resource revenues.

The SSA picture is different. Within SSA, both slices produce large and statistically significant coefficients (0.116 excluding mineral-rent countries, 0.139 in mineral-rent SSA), so the mineral-rent split does not account for the SSA concentration; SSA-specific factors (institutional, demographic, or relating to spell length) carry independent weight. Outside SSA, the activity effect is small and imprecise in both slices. We therefore read the mineral-rent split as one supporting strand of evidence for the composition channel at the global level rather than as the explanation for the SSA pattern. The

Table 7. Democracy heterogeneity (2005–2017)

	(1) ln NTL	(2) NO ₂ int.	(3) PM _{2.5} int.
Low democracy	0.066* (0.029)	−0.081* (0.038)	−0.062* (0.029)
High democracy	0.042* (0.018)	−0.021 (0.023)	−0.036* (0.018)
<i>N</i> (low / high)	203k / 408k	185k / 391k	202k / 407k

Sample split at the median country-mean V-Dem electoral democracy score (0.499) over 2005–2017 (11). Each cell is a separate regression with ADM2 fixed effects, country-year fixed effects, and leader-spell clustered standard errors. * $p < 0.05$.

Table 8. Robustness to country-level clustering

Spec.	Coef.	Spell SE	Ctry SE	Ctry <i>p</i>
NL, 1992–2013	0.043	0.014	0.018	0.016
NL, 2005–2013	0.033	0.014	0.017	0.053
NL, 2005–2017	0.049	0.015	0.020	0.013
NO ₂ int., 2005–2017	−0.039	0.019	0.025	0.119
PM _{2.5} int., 2005–2017	−0.044	0.015	0.020	0.025

NL denotes nightlights. Baseline standard errors cluster by leader spell; the robustness column clusters by country, following the original Hodler–Raschky convention.

WDI mineral-rent variable is country-level and slow-moving, so the split absorbs other features of resource-rich states.

Democracy and accountability. Table 7 splits the sample at the median country-mean V-Dem electoral democracy score (0.499) over 2005–2017. The nightlights favoritism effect is present in both groups, but the green favoritism pattern is concentrated in low-democracy countries. The NO₂ intensity coefficient in high-democracy countries is small and insignificant (−0.021, $p = 0.35$), while in low-democracy countries it is −0.081 ($p = 0.031$). The PM_{2.5} intensity result is significant in both groups but roughly twice as large in low-democracy contexts.

This pattern is consistent with a discretion mechanism: in settings with weaker accountability, leaders have more latitude to determine not just how much investment reaches their birth region but what kind. In more democratic contexts, birth regions still receive more activity but the composition is less selective.

Robustness. Table 8 shows that the main results survive country-level clustering. The nightlights coefficient in the harmonized window remains significant ($p = 0.013$). The PM_{2.5} intensity result also survives ($p = 0.025$), while NO₂ intensity loses precision ($p = 0.119$). PM_{2.5} thus provides the more conservative evidence for green favoritism.

Limitations. Several caveats apply. First, satellite NO₂ and PM_{2.5} are vertically integrated retrievals translated to surface concentrations by chemical transport models; they are noisy proxies for ground-level exposure and are known to perform less well over bright surfaces (deserts, snow) and over very dark, low-emission regions (9). Measurement error of this kind would generally attenuate coefficients toward zero rather than generate spurious negatives. Second, nightlights are an imperfect proxy for economic activity, especially at low luminance and in agricultural regions where output need not translate into emitted light. The harmonized DMSP-to-VIIRS series

mitigates the saturation issues of raw DMSP (8) but does not eliminate them. Third, birthplace coding in PLAD relies on biographical sources of varying quality; misclassification of a leader’s birth region would attenuate the treatment indicator. Fourth, the PM_{2.5} result is statistically more robust than the NO₂ result under country-level clustering, and we therefore treat PM_{2.5} as the more conservative evidence for green favoritism while NO₂ provides the cleaner pre-trends in the event study; the two indicators are complementary rather than interchangeable. Finally, the paper documents a pattern but cannot isolate the underlying channel. Composition, enforcement, and sorting all remain consistent with the aggregate evidence, and distinguishing them requires project-level or firm-level data not available at the global panel scale used here.

Discussion

This paper asked whether birth regions of national leaders receive disproportionately clean economic growth. The answer is yes. Birth-region favoritism in nightlights is a durable feature of the data, present across sensor generations, sample windows, and clustering schemes. The new result is that this favoritism is environmentally selective: birth regions grow brighter but not proportionally dirtier. Negative pollution-intensity coefficients, flat NO₂ pre-trends in the event study, and the SSA concentration all point to birth-region growth that differs in composition from growth elsewhere. The mineral-rent decomposition supplies one strand of mechanism evidence: globally, the activity effect is about three times larger in mineral-rent economies than outside them, consistent with discretionary resource revenues funding the kind of large, observable, low-emission public works (ports, airports, roads, government facilities) that the Hambantota case illustrates. The mineral-rent split does not, however, account for the Sub-Saharan concentration: within SSA, the activity effect is large and significant in both mineral-rent and non-mineral slices, so SSA-specific factors carry independent weight. The data do not let us cleanly isolate composition from related channels such as enforcement and sectoral sorting, but the joint pattern of green favoritism, the SSA concentration, and the mineral-rent gating at the global level is what one would expect if leaders in fiscally permissive settings direct resource-funded infrastructure rather than industrial activity toward home districts.

The finding has implications for environmental inequality. Controlling for activity level, districts without a sitting leader receive relatively more of the pollution-intensive growth. Future work could test whether specific project categories—port and airport construction, electrification, road building—account for the clean-intensity pattern, by linking nightlights and pollution outcomes to administrative investment records.

Materials and Methods

Data sources and panel construction are described in Section 1. The nightlights panels are built at ADM2 resolution by extracting annual zonal means from DMSP DN rasters (1992–2013) and harmonized DMSP-to-VIIRS rasters (2005–2017) using GADM level-2 boundaries. NO₂ and PM_{2.5} are extracted from ACAG annual surface-concentration grids using the same boundaries. Leader birthplace districts are coded from PLAD and matched to GADM

ADM2 polygons. The democracy split uses country-mean V-Dem electoral democracy scores (`v2x_polyarchy`) over 2005–2017 (11), with countries split at the cross-country median (0.499). All regressions use `linearmodels` PanelOLS with absorbed fixed effects; standard errors are clustered by leader spell unless otherwise noted.

Data, Materials, and Code Availability

Code and manuscript sources are available at <https://github.com/RichSchulz/political-leader-pollution>. External inputs include PLAD (7), GADM (10), ACAG NO₂ (9) and PM_{2.5}, Li et al. harmonized nightlights, and DMSP annual composites obtained under the providers' access conditions.

ACKNOWLEDGMENTS. We thank Professor Joshua Blumenstock for feedback. All errors are our own.

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